

Pierluigi Rinaldi

H-index = 34. Citations: 3587. Source: NASA ADS.

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🌐 <https://www.stsci.edu/stsci-research/research-directory/pierluigi-rinaldi>

🌐 <https://pierluigi-rinaldi.github.io/>

🌐 <https://www.linkedin.com/in/pierluigi-rinaldi-2276b9192/>

Research Interests

- Galaxy formation and evolution at intermediate and high redshifts; stellar mass assembly; star formation activity and its evolution over cosmic time; infrared-selected galaxies; multi-wavelength photometric and spectroscopic follow-up; Little Red Dots, active galactic nuclei and supermassive black hole growth; galaxy-SMBH co-evolution across cosmic time; (proto)clusters and environmental effects on galaxy evolution.
- Instrumentation and Software developing.

Higher Education

- 2020 – 2024 **Ph.D. Astronomy, University of Groningen.**
Thesis title: *Unfolding the early Universe with JWST*
Supervisor(s): Karina I. Caputi, Filippo Fraternali.
- 2017 – 2020 **M.Sc. Astrophysics and Cosmology (cum laude), Alma Mater Studiorum - University of Bologna** in Galaxy formation and evolution.
Thesis title: *Simulation of high-redshift galaxy spectra in the Euclid Deep Fields.*
Supervisor(s): Andrea Cimatti, Michele E. M. Moresco.
- 2012 – 2017 **B.Sc. Physics, University of Bari Aldo Moro** in Planetary Formation.
Thesis title: *The condensation: from clouds to protoplanetary systems.*
Supervisor(s): Savino Longo.

Appointments (after PhD)

- 2026 – 2029 **Cosmic Frontier Center Prize Fellow** – Cosmic Frontier Center, The University of Texas at Austin, Austin (TX), USA.
- 2025 – 2026 **Postdoctoral Research Fellow** (funded through PI-led JWST grant) – Space Telescope Science Institute (STScI) in Baltimore (MD), USA.
- 2024 – 2025 **Postdoctoral Research Associate** – Department of Astronomy and Steward Observatory, Tucson (AZ), USA. Mentors: George H. Rieke & Stacey Alberts

Teaching

- 2021 – 2024  **Teaching Assistant.** Astrophysical Hydrodynamics, Bachelor course, Science and Engineering, Rijksuniversiteit Groningen.
- 2019 – 2020  **Teaching Assistant.** Physics, First cycle degree programme (L) in Biological Sciences (cod. 8012), Biological Sciences, Alma Mater Studio-rum - University of Bologna.

Supervision and Mentorship

- 2026–2030  **PhD Student** – Yifan Wang at Rijksuniversiteit Groningen. [Co-Supervisor]
- 2026  **PhD Student** – Michele Catone at University of Padova. [Mentor]
-  **Master Student** – Gaurav Senthil Kumar at University of Stockholm. [Mentor]
- 2024  **Bachelor Student** – Nadav Peleg-Brochestein at Rijksuniversiteit Groningen. [Co-Supervisor]
- 2022 – 2026  **PhD Student** - Rafael Navarro-Carrera at Rijksuniversiteit Groningen. [Mentor]

Professional expertise

- Coding  Python, L^AT_EX, C++.
- Data reduction  Reduction of JWST data (MIRI, NIRCcam, and NIRISS imaging, NIRCcam/WFSS, and NIRSpec).
- Data Analysis  UV/Optical (HST) and NIR/MIR (JWST) imaging and spectroscopic data, Spectral Energy Distribution (SED) fitting, photometry, multi-wavelength catalogues, and spectral analysis.
- Software  JWST PIPELINE for MIRI, NIRCcam, NIRSpec, and NIRISS, BAGPIPES, PROSPECTOR, LEPHARE, EAZY, GRIZLI, MSAEXP MIRISIM (to simulate MIRI data) MIRAGE (to simulate JWST's NIRCcam, NIRISS, and FGS instruments), SOURCEEXTRACTOR (SEXTRACTOR).

Professional Membership & Collaborations

- 2026 – now  Member of the Steering Committee of the JADES Federation.
-  Official member of MIRAGE
-  Official member of the Roman eXtreme Deep Field (RXDF) team
-  Official member of DIVER & EMERALD
- 2024 – now  PI with Edoardo Iani of the MIRIC team (GO 5578 – leader of MIRI data reduction)
-  Member of the Nancy Grace Roman Telescope Working Groups (software, simulations, synergies, and calibration)
-  Official member of the PRISMS team.

Professional Membership & Collaborations (continued)

- Official member of the OASIS team.
- Official member of the SMILES team.
- Official member of the JADES team.
- Official member of the WST team.
- 2021 – now ■ Official (key) member of the MIRI (EU and US) team.
- Official (key) member of MIDIS team – NIRCcam and MIRI data reduction.
- 2019 – 2021 ■ Member of the Euclid Consortium.

Service to the astronomical community

- 2026 ■ ALMA Reviewer for the Cycle 13
- Reviewer for the Space Astronomy Summer Program for STScI
- 2025 ■ JWST TAC Panel Support for JWST Cy5
- 2020 – now ■ Reviewer for Scientific Journals: AAS journals, Nature, Nature astronomy, MNRAS

Personal Awards and Funding

- 2026 ■ Juan de la Cierva (JdIC) Fellowship, Centro de Astrobiología (CAB), Madrid, Spain (awarded as the top-ranked candidate) $\approx$ \$153,000
- \$220,000 for JWST Cycle 5 proposal GO 10311
- Cosmic Frontier Center Prize Fellowship, UT Austin, USA $\approx$ \$270,000
- Center for Astrophysics and Space Science (CASS) Prize Fellowship, NYUAD (UAE) $\approx$ \$300,000 [declined]
- 2024 ■ \$222,000 for JWST Cycle 3 proposal GO 5578
- [Royal Astronomical Society Group Achievement Award for JWST-MIRI Team](#)

Telescope time

- Summary ■ As a PI $\approx$ 100h ($\approx$ \$440,000); as a Co-I $>$ 1500h ($>$ \$3M)
- PI ■ **“Slicing the Saguaro: Dissecting a Little Red Dot Descendant at Cosmic Noon with NIRSpec/IFU”** (JWST cycle 5; PID: 10311) [27h]
- **“The MIRI deep imaging survey of the lensing clusters Abell2744 and MACS0416”** (JWST cycle 3; PID: 5578) [75h]
- Co-I — Roman Cycle 1 ■ PID: 19009, 19049, 19017, 19100, 19098, 19062, 2001

Telescope time (continued)

- Co-I — JWST Cycle 5  PID: 12340, 10790, 10311, 10202, 10149
- Co-I — JWST Cycle 4  PID: 10830, 8544, 8051, 8018, 7935, 7605
- Co-I — JWST Cycle 3  PID: 6511, 5578
- Co-I — JWST Cycle 2  PID: 3538, 2926, 3432
- Co-I — JWST Cycle 1  PID: 1283
- Co-I — JWST DDT  PID: 9433
- Co-I — ALMA Cycle 12  2025.1.01067.S, 2025.1.00147.S

Conferences, Seminars, and Workshops (selected)

- 2026  **Invited Talk**/Online Workshop on Little Red Dots
-  **Contributed Talk**/Conference – The Drivers of Galaxy Evolution through Cosmic Time (Florence, IT)
-  **Invited Talk**/Consortium Meeting – Final MIRI European Consortium meeting; Plenary talk on the *Impact of MIRI on the high-z Universe* (Edinburgh, SCT)
-  **Contributed Talk**/Conference – Towards the Full Bloom of First Star, Galaxy, and Black Hole Formation Exploration (Tokyo, JP)
-  **Invited talk**/Workshop – The First Billion Years: 5 Question in 5 Days (Aspen, US)
- 2025  **Invited talk**/Seminar – Institute of Science and Technology Austria (ISTA) (Vienna, AT)
-  **Invited Talk**/Seminar – Kavli Institute for Cosmology (KICC) (Cambridge, UK)
-  **Contributed Talk**/Conference – First Galaxies (Oxford, UK)
-  **Invited Talk**/Consortium Meeting – The MIRI EU meeting (Madrid, Spain)
-  **Invited Talk**/Consortium Meeting – The NIRCам meeting (Biosphere 2, Tucson, US)
- 2024  **Invited Talk**/Consortium Meeting – The MIRI US and EU meeting (Biosphere 2, Tucson, US)
-  **Invited Talk**/Seminar – Scuola Internazionale Superiore di Studi Avanzati (Trieste, IT)
-  **Invited talk**/Seminar – Space Telescope Science Institute (Baltimore, US)
- 2023  **Contributed Talk**/Conference – Netherlands Research School for Astronomy (Leiden, NL)

Conferences, Seminars, and Workshops (selected) (continued)

- **Invited Talk**/Consortium Meeting – MIRI European Consortium meeting. Plenary talk on the *Highlights of the MIRIM science in the Hubble eXtreme Deep Field* (Dublin, IRE).
- **Contributed Talk**/Conference – The James Webb Space Telescope turns one: the birth and growth of galaxies (Sexton Center for Astrophysics, Sesto, IT).
- **Invited talk**/Conference – Shedding new light on the first billion years of the Universe (Marseille, FR).
- 2022 ■ **Invited talk**/Seminar – INAF/OAS Padova (Padova, IT).
- **Invited talk**/Seminar – Astrophysics Talks at INAF/OAS Bologna (Bologna, IT).
- 2021 ■ **Contributed Talk**/Conference – Netherlands Research School for Astronomy (online).

Outreach, public engagement, and press

- 2026 ■ [NASA Webb Explores Family Tree of Newly Discovered Distant Objects](#) – interviewed by the NASA Webb Mission Team (July 29th)
- [Notti d'estate ad Arcetri](#) – Public lecture on galaxy formation and evolution across cosmic time (June 9th).
- [Aurora Astronomy Seminar Series at UCR](#) (May 4th).
- [Virgil: a black hole that should not exist](#) (BBC Sky at Night Magazine; March 2026 issue) – Interviewed by Anita Chandran (January 9th)
- [The Birth of Light](#) – Interviewed by Rebecca Boyle (January 8th)
- [Weird 'Little Red Dots' in space are something we've never seen](#) (Scientific American) – Interviewed by Rebecca Boyle (January 8th)
- 2025 ■ [Talk on Little Red Dots with undergraduate students at Bates College](#) – invited by Aleks Diamond-Stanic (December 18th)
- [The monster hiding in plain sight: JWST reveals cosmic shapeshifter in the early universe](#)
- [Little Red Dots frequently reveal clumpy morphologies, and some of these also bear the hallmarks of active black holes](#)
- 2024 ■ [Talk at Associazione Astronomiamo, Milan \(IT\) – Beyond Hubble: Unfolding the Early Universe with JWST](#) (April 18th).
- 2023 ■ [MIRI instrument on JWST detects H-alpha emission during the Epoch of Reionization for the first time.](#)
- [The James Webb telescope detects much more light than expected coming from the primaeval universe.](#)
- 2022 ■ [Zenit](#) (Dutch Astronomical Magazine), December 2022 issue — Article on starburst galaxies
- [Astronomers see the infrared switch on in young universe.](#)
- [Early Universe bristled with starburst galaxies.](#)
- [Talk – Kapteyn Science Day](#) (December 9th).
- [Member of the Astronomy on Tap in Groningen.](#)

Conference and event organization

- 2023  LOC – 6th NOVA evaluation, Groningen (January 23th - 25th).
2022  LOC – JWST First Data Release - Public Presentation in Groningen, DOT Planetarium (July 12th).

Visting

- 2026  Kapteyn Astronomical Institute, Groningen (NL) (May 22nd - June 6th)
2025  Institute of Science and Technology Austria (ISTA), Vienna (December 8th - 11th)
  Kapteyn Astronomical Institute, Groningen (NL) (December 1st - 5th)
  Kavli Institute of Cosmology, Cambridge (April 14th - 17th)
2024  Kapteyn Astronomical Institute, Groningen (NL) (December 8th - 15th)
  Scuola Internazionale Superiore di Studi Avanzati, Trieste (May 25th - 29th)
  Royal Swedish Academy of Sciences, Stockholm (May 20th - 24th)
  Space Telescope Science Institute (STScI), Baltimore (January 16th - 19th)
2023  Royal Irish Academy, Dublin (September 25th - 29th).
  Katholieke Universiteit Leuven, Leuven (February, 13th -17th).
2022  Universidad Complutense de Madrid, Madrid (June 6th - 10th).
  INAF OAS Bologna (Bologna) & INAF - Osservatorio Astronomico di Padova (Padova) (May 23th - 24th).
2021  Centro de Astrobiología (CAB), Madrid (November 10th - 14th).

Publications (First and Second author)

Mentored papers are underlined

- **P. Rinaldi**, K. Hainline et al. 2026 — <https://ui.adsabs.harvard.edu/abs/2026arXiv260407138R/abstract>
- E. Iani, **P. Rinaldi** et al. 2026 — <https://ui.adsabs.harvard.edu/abs/2026arXiv260317967I/abstract>
- **P. Rinaldi**, S. Alberts et al. 2026 — <https://ui.adsabs.harvard.edu/abs/2026arXiv260224162R/abstract>
- **P. Rinaldi**, G. H. Rieke et al. 2025 — <https://ui.adsabs.harvard.edu/abs/2025arXiv250717738R/abstract>
- **P. Rinaldi**, P. G. Pérez-González et al. 2025 — <https://ui.adsabs.harvard.edu/abs/2025arXiv250401852R/abstract>
- **P. Rinaldi**, N. Bonaventura et al. 2025 — <https://ui.adsabs.harvard.edu/abs/2024arXiv241114383R/abstract>
- **P. Rinaldi**, R. Navarro-Carrera et al. 2025 — <https://ui.adsabs.harvard.edu/abs/2024arXiv240613554R/abstract>
- R. Navarro-Carrera, **P. Rinaldi** et al. 2025 — <https://ui.adsabs.harvard.edu/abs/2024arXiv241023249N/abstract>

Publications (First and Second author) (continued)

- E. Iani, **P. Rinaldi** et al. 2025 — <https://ui.adsabs.harvard.edu/abs/2024arXiv240618207I/abstract>
- P. G. Pérez-González, **P. Rinaldi** et al. 2024 — <https://ui.adsabs.harvard.edu/abs/2024arXiv240216942P/abstract>
- K. I. Caputi, **P. Rinaldi** et al. 2024 — <https://ui.adsabs.harvard.edu/abs/2023arXiv231112691C/abstract>
- **P. Rinaldi**, K. I. Caputi et al. 2024 — <https://ui.adsabs.harvard.edu/abs/2023arXiv230915671R/abstract>
- R. Navarro-Carrera, **P. Rinaldi** et al. 2023 — <https://ui.adsabs.harvard.edu/abs/2023arXiv230516141N/abstract>
- **P. Rinaldi**, K. I. Caputi et al. 2023 — <https://ui.adsabs.harvard.edu/abs/2023ApJ...952..143R/abstract>
- **P. Rinaldi**, K. I. Caputi et al. 2022 — <https://ui.adsabs.harvard.edu/abs/2022ApJ...930..128R/abstract>